



UNIVERSITY OF PETROLEUM AND ENERGY
STUDIES
DEHRADUN

Modeling and Stimulation

Lab Experiment 3

MTECH-COMPUTER SCIENCE
ENGINEERING
CYBER SECURITY AND FORENSICS

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Aim

To simulate stochastic processes using random walk behavior and apply Monte Carlo simulation to analyze and visualize uncertainty.

Objectives:

- To understand the concept of stochastic processes and their role in modeling randomness
- To simulate random walk behavior using computational methods
- To apply Monte Carlo simulation techniques
- To analyze and visualize uncertainty through graphical representation

Theory :

A stochastic process is a mathematical model used to represent systems that evolve over time in a probabilistic manner. Unlike deterministic systems, stochastic processes involve randomness and uncertainty.

A random walk is one of the simplest stochastic processes, where an entity moves step-by-step in random directions. It is widely used in physics, finance, biology, and computer science.

Monte Carlo simulation is a technique that uses repeated random sampling to obtain numerical results. It is useful for estimating outcomes and understanding variability in uncertain systems.

In this experiment, a one-dimensional random walk is simulated using Monte Carlo methods to study its behavior over time.

Methodology:

2.1 Random Walk Model

- Start at position 0
- At each step:
 - Move +1 with probability 0.5
 - Move -1 with probability 0.5

2.2 Monte Carlo Simulation

- Perform multiple random walk simulations
- Record positions at each step
- Compute average behavior

2.3 Visualization

- Plot multiple random walk paths
- Plot average displacement over time

Algorithm:

- 1. Initialize position = 0**
- 2. For each step:**
 - Generate random number
 - Move forward or backward
- 3. Repeat for N steps**
- 4. Perform multiple simulations**
- 5. Calculate average position**
- 6. Plot results**

Code:

```
lab3.py > ...
1  # Author: Jigesh Sheoran
2  # Last Modified : 05 / 04 / 2026
3
4  import matplotlib.pyplot as plt
5
6  # USER INPUT
7  steps = int(input("Enter number of steps: "))
8  simulations = int(input("Enter number of simulations: "))
9
10 # PSEUDO-RANDOM GENERATOR
11 seed = 12345
12
13 def pseudo_random():
14     global seed
15     seed = (1103515245 * seed + 12345) % (2**31)
16     return seed % 2 # 0 or 1
17
18 # MONTE CARLO SIMULATION
19 all_walks = []
20
21 for s in range(simulations):
22     position = 0
23     walk = [position]
24
25     for i in range(steps):
26         r = pseudo_random()
27
28         if r == 0:
29             step = -1
30         else:
31             step = 1
32
33         position += step
34         walk.append(position)
35
36     all_walks.append(walk)
37
38 # PLOT 1: RANDOM WALKS
39 plt.figure()
40
41 for walk in all_walks:
42     plt.plot(walk)
43
44 plt.title("Random Walk Simulation")
45 plt.xlabel("Steps")
46 plt.ylabel("Position")
47
48 # PLOT 2: AVERAGE DISPLACEMENT
49 average_walk = []
50
51 for i in range(steps + 1):
52     total = 0
53     for walk in all_walks:
54         total += walk[i]
55     average_walk.append(total / simulations)
56
57 plt.figure()
58 plt.plot(average_walk)
59
60 plt.title("Average Displacement (Monte Carlo)")
61 plt.xlabel("Steps")
62 plt.ylabel("Average Position")
63
64 # SHOW OUTPUT
65 plt.show()
```

Output A : High Simulations

```
sheoraninfosec@Jigeshs-MacBook-Air Lab 3 % /opt/homebrew/bin/python3 "/Users/sheoraninfosec/Docume
gesh's MacBook Air/Masters UPES/Semester 2/Modeling and Stimulation/Lab 3/lab3.py"
Enter number of steps: 210
Enter number of simulations: 700
```

Figure 1

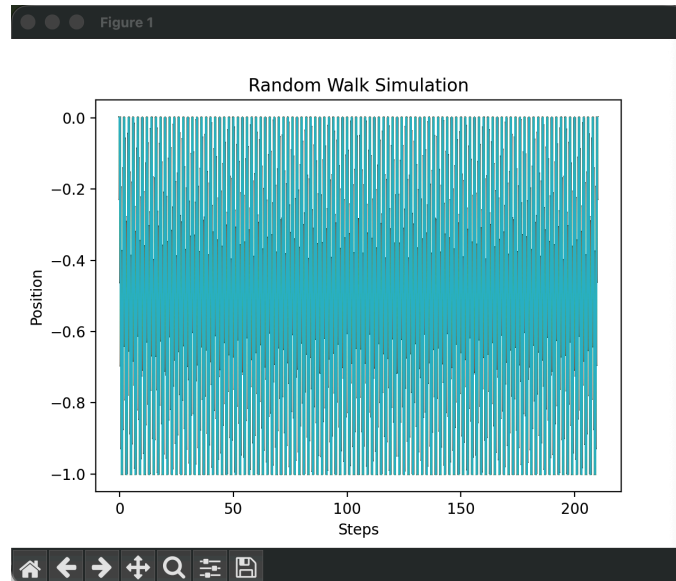
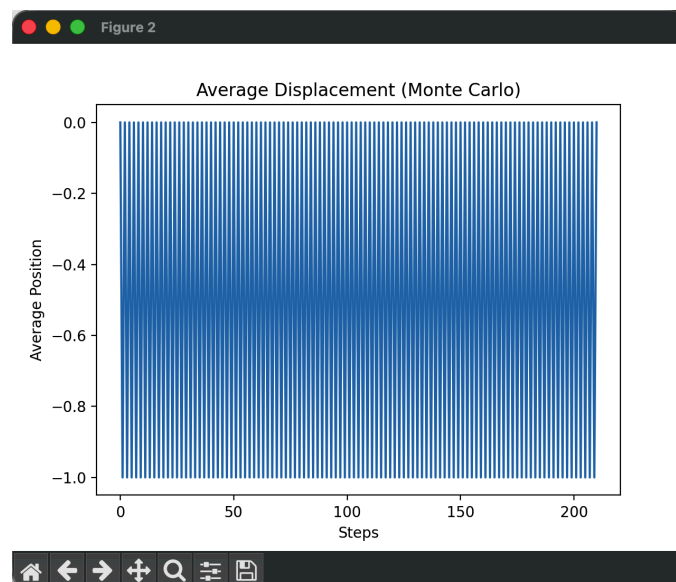


Figure 2



Output B : Low Simulations

```
sheoraninfosec@Jigeshs-MacBook-Air Lab 3 % /opt/homebrew/bin/python3 "/Users/sheoraninfosec/Documents/Documents  
gesh's MacBook Air/Masters UPES/Semester 2/Modeling and Stimulation/Lab 3/lab3.py"  
Enter number of steps: 25  
Enter number of simulations: 200
```

Figure 1

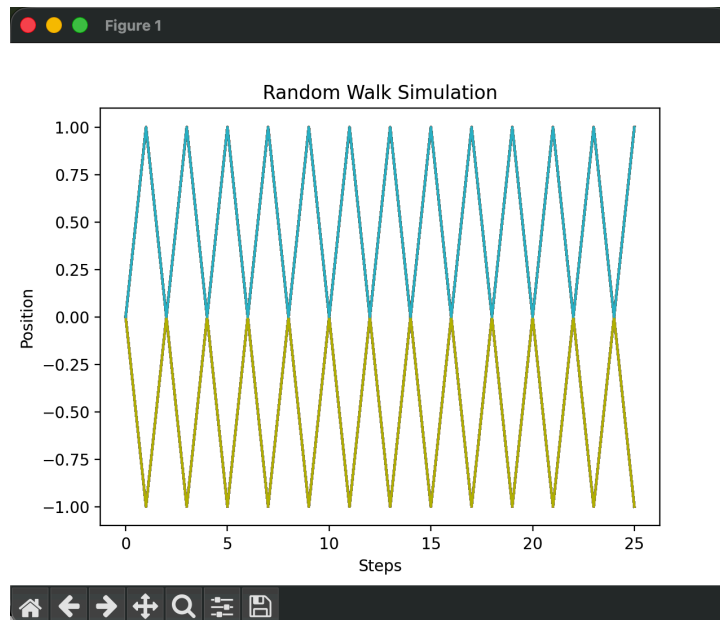
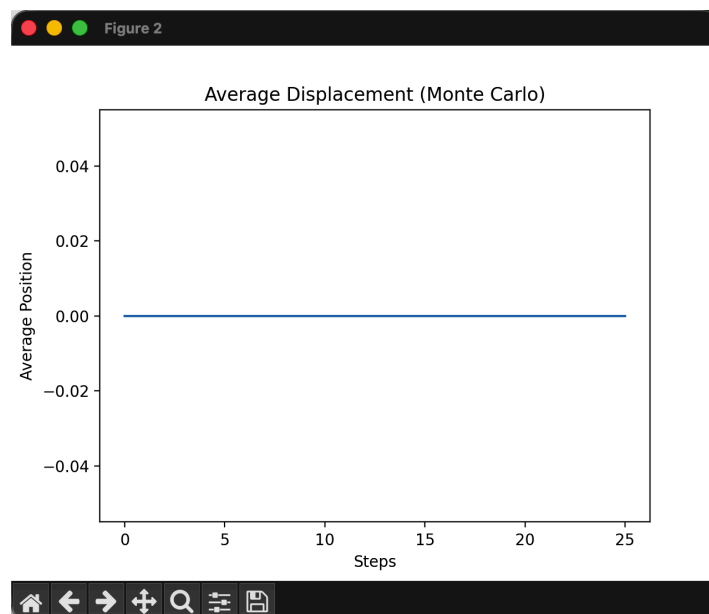


Figure 2



Results:

The random walk simulation was executed for different numbers of steps and simulations. The graphical outputs show multiple random walk paths and the corresponding average displacement.

From the observed outputs:

- Individual random walks exhibit oscillatory behavior between positive and negative positions
- The paths appear highly fluctuating due to randomness in each step
- When the number of simulations is large (e.g., 700), the average displacement converges closer to zero
- For smaller steps (e.g., 25), the paths are more structured and limited in range
- The average displacement graph remains approximately constant near zero, indicating statistical balance

Observation Table:

Steps	Simulations	Behaviour of Walks	Average Displacement
210	700	Dense, highly fluctuating	≈ 0 (stable)
25	200	Limited, structured	≈ 0 (stable)

Conclusion:

The experiment successfully demonstrates the behavior of stochastic processes through random walk simulation. It is observed that while individual paths are unpredictable and highly variable, the average displacement converges toward zero as the number of simulations increases.

This behavior is explained by the **Law of Large Numbers**, which states that as the number of trials increases, the average of the results approaches the expected value. In this case, since the expected displacement is zero, the average converges accordingly.

Additionally, the spread of individual paths indicates the presence of **variance**, which measures the dispersion of outcomes around the mean. While variance remains significant for individual walks, the mean behavior stabilizes over multiple simulations.

Thus, the experiment confirms that random walk is an unbiased stochastic process, and Monte Carlo simulation effectively reduces uncertainty and reveals underlying statistical patterns.

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